performance as well as maintenance cost is increased. In such cases, the use of a sleeve or liner can restore proper clearances to an engine. Due to prolonged use, the liner will wear and it can be replaced at lower cost. Cylinder liners are of two types namely dry liner and wet liner.

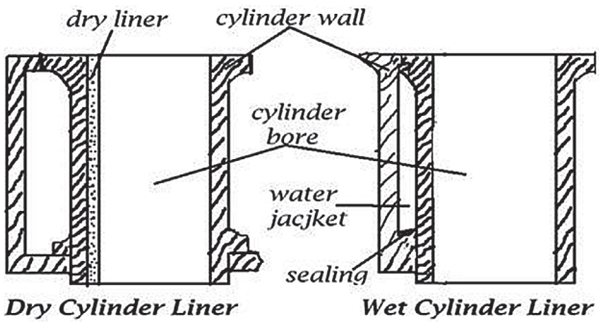


Figure 5.3.2 Dry and Wet Liner

(a) Dry liner

The outer diameter of liner and inner diameter of the cylinder wall are manufactured to fit perfectly. Liners are assembled with the help of hydraulic press or by cooling the liner. The dry type liner is not in direct contact with the coolant hence the name "Dry". The cylinder liner is surrounded by the cylinder body. The cylinder body is contacted by the cooling water so as to achieve heat dissipation. Liner thickness would be 1.5 mm to 3.0 mm. Flanges and seals are at top surface of cylinder.



Figure 5.3.2(a) Dry Liner

Wet liners may have a cooling water space between the engine block and liner, or they may have integral cooling passages. Liners with integral cooling passages are sometimes referred to as water-jacket liners.

(b) Wet liner

The inner surface of wet liner are perfectly machined and honed. The outer surface is not machined. They are referred to as "wet liners" because their outer sides come in direct contact with the engine's coolant. The thickness of the liner will be 3.0mm to 6.0 mm. It is sealed by a metallic sealing ring from top and a rubber sealing ring at the bottom. A wet liner cylinder block features cylinder walls that are entirely removable, which fit into the block by means of special gaskets.



Figure 5.3.2(b) Wet Liner

5.3.3 *Cylinder head*

In an engine, the cylinder head assembled above the cylinders on top of the cylinder block with the help of studs. It closes in the top of the cylinder, forming the combustion chamber. This joint is sealed by a head gasket. The head also provides passage for intake of air and fuel and exhaust of burnt gases, water cooling passage. The head can also be a place to mount the valves, spark

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